

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

From coral reefs to climate engineering and space junk to bacteriophages, it's all here!



Musk's Space Roadster may come home

Researchers have worked out the likely future landing spots of Elon Musk's Space Roadster. There is a six percent chance that it will come back to the Earth, and a 2 percent chance that it will end up on Venus. There is also a remote chance that it could strike Mars, or fly out to the Kuiper belt.

<http://dgit.in/space-roadster>

WHAT'S NEW

NASA finds mother lode of Cretaceous animal footprints

N

ASA is usually known for studying the depths of space, but stumbled upon something from the dinosaur age.

NASA has carefully extracted and studied a slab of rock discovered in its Goddard Space Flight Center in Greenbelt, Maryland. The slab of rock, about two and a half meters long and one meter wide, has almost 70 well preserved tracks from eight species. Impressions by nodosaurs, sauropods, theropods and pterosaurs share space with 26 mammalian tracks on the slab. The imprints of all the tracks are believed to have been left behind by the animals between 110 to 112 million years ago, within the span of a few



days. The location is believed to have been at the fringes of a wetland. The find has the the largest mammalian footprint from that time period. The creature that left it behind was a racoon sized animal, with a hand that had five digits, very similar to the human hand today.

<http://dgit.in/NASAfossil>

Researchers teach tricks to a worm uploaded to a computer

The nematode *Caenorhabditis elegans* is one of the simplest microanimals around. *C. elegans* has the distinction of being the only animal whose entire connectome has been uploaded to a computer. A connectome is a map of all the neural connections in the nervous system of an organism, which can be thought of like



a wiring diagram. This is not a computer program, it is an actual digital copy of the worm. Even with the simple nervous system, the tiny worm can navigate, eat bacteria and

respond to touch. If the worm feels a touch, it wriggles away. The digital version of the worm reacts to a simulated touch in the same way as the real nematode would, simply because the response is hard coded in the neural network of the organism. That response was used to teach the worm a new trick – to balance a pole.

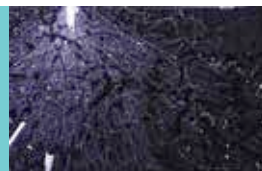
<http://dgit.in/poleworm>



Scientists demonstrate stopping electrons with light

Researchers fired high powered laser pulses at a high energy electron beam, to demonstrate for the first time that electrons can be stopped with light. The conditions created were similar to the surroundings of black holes and quasars.

<http://dgit.in/astragemini>



TIFR researchers study electrons travelling faster than light

The setup involved firing a laser at a glass target. Since light slowed down within the glass, the electrons moved faster than light within the medium, emitting measureable Cherenkov radiation in the process.

<http://dgit.in/tifrftl>



Planets in the TRAPPIST-1 system could have more water than the Earth

New measurements of the Planets in the TRAPPIST-1 system suggests some of the planets may have as much as 250 times the water as the oceans of Earth.

<http://dgit.in/trappistwater>